

Problem 16.14

Which one is greater, $s_{\overline{n}} - a_{\overline{n}}$ or $\ddot{s}_{\overline{n}} - \ddot{a}_{\overline{n}}$?

Solution

First compare the two expressions algebraically. We know that

$$\ddot{s}_{\overline{n}} = (1+i)s_{\overline{n}} \quad \text{and} \quad \ddot{a}_{\overline{n}} = (1+i)a_{\overline{n}}.$$

Therefore,

$$\begin{aligned} \ddot{s}_{\overline{n}} - \ddot{a}_{\overline{n}} &= (1+i)s_{\overline{n}} - (1+i)a_{\overline{n}} \\ &= (1+i)(s_{\overline{n}} - a_{\overline{n}}). \end{aligned}$$

Since $i > 0$, we have

$$1+i > 1.$$

Also, for a positive interest rate,

$$s_{\overline{n}} > a_{\overline{n}},$$

so

$$s_{\overline{n}} - a_{\overline{n}} > 0.$$

Multiplying this positive quantity by $1+i > 1$ makes it larger. Hence,

$$\ddot{s}_{\overline{n}} - \ddot{a}_{\overline{n}} > s_{\overline{n}} - a_{\overline{n}}.$$

Equivalently, using formulas,

$$\begin{aligned} s_{\overline{n}} - a_{\overline{n}} &= \frac{(1+i)^n - 1}{i} - \frac{1 - (1+i)^{-n}}{i} \\ &= \frac{(1+i)^n + (1+i)^{-n} - 2}{i}, \end{aligned}$$

while

$$\begin{aligned} \ddot{s}_{\overline{n}} - \ddot{a}_{\overline{n}} &= \frac{(1+i)^n + (1+i)^{-n} - 2}{i(1+i)^{-1}} \\ &= (1+i)(s_{\overline{n}} - a_{\overline{n}}). \end{aligned}$$

Therefore, the greater quantity is

$$\boxed{\ddot{s}_{\overline{n}} - \ddot{a}_{\overline{n}}}.$$